system #2

$$\begin{cases} 1\cdot x+\frac{6}{4}\cdot y=1 & A_2=\frac{1}{4}\cdot A_1 \\ 3\cdot x+7\cdot y=13 & B_2=B_1 \end{cases}$$

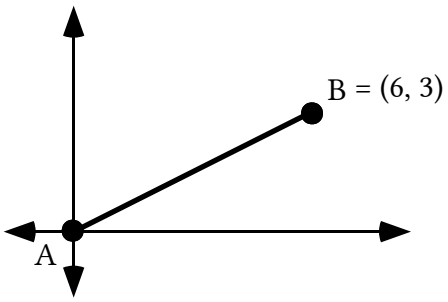
system #3

$$\begin{cases} 1\cdot x+\frac{6}{4}\cdot y=1 & A_3=A_2 \\ 0\cdot x+\frac{10}{4}\cdot y=10 & B_3=-3\cdot A_2+B_2 \end{cases}$$

comes from $-3\cdot(1\cdot x)+3\cdot x$

comes from $-3\cdot\left(\frac{6}{4}\cdot y\right)+7\cdot y$

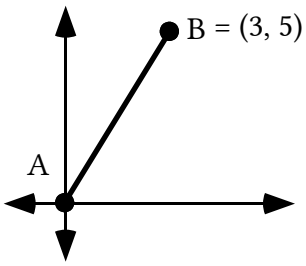
comes from $-3\cdot(1)+13$


B = (6, 3)
A

$$\begin{cases} 4 \cdot x + 2 \cdot y = 2 & A_1 \\ 4 \cdot x + \overline{1} \cdot y = 11 & B_1 \end{cases}$$

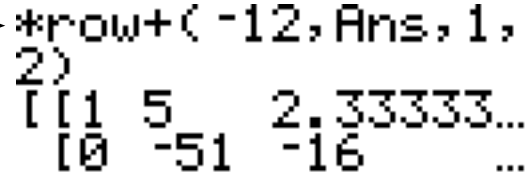
$$\begin{cases} A_2 = \frac{1}{4} \cdot A_1 \\ B_2 = B_1 \end{cases}$$

$$\begin{cases} A_3 = A_2 \\ B_3 = \bar{4} \cdot A_2 + B_2 \end{cases}$$


B = (3, 5)
A

```
augment([A],[C])
     [[3   15 7  ]
      [12  9   12]]
```

command to implement the step coded by $B_3 = -12 \cdot A_2 + B_2$



```
[[1 5 2.3333333…
 [0 1 .31372549…
*row+(-5,Ans,2,1
)
[[1 0 .76470588…
 [0 1 .31372549…
```

```
Ans▸Frac
    [[1  0  13/17]
     [0  1  16/51]]
```

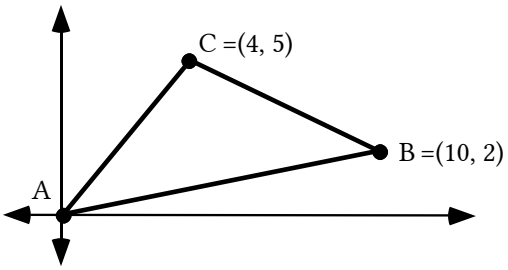
$$\left[\begin{array}{cc|c} 3 & 1 & 1 \\ -2 & -3 & 4 \end{array}\right] \begin{matrix} A_1 \\ B_1 \end{matrix}$$

$$\left[\begin{array}{c|c} & \\ & \end{array}\right] \begin{array}{l} A_2 = \frac{1}{3} \cdot A_1 \\ B_2 = B_1 \end{array}$$

$$\left[\begin{array}{cc|c} 14 & 21 & -17 \\ 20 & 6 & 8 \end{array}\right] \begin{matrix} A_1 \\ B_1 \end{matrix}$$

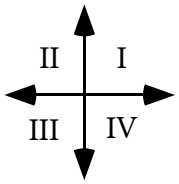
$$\left[\begin{array}{c|c} & \\ & \end{array}\right] \begin{array}{l} A_3 = A_2 \\ B_3 = -20 \cdot A_2 + B_2 \end{array}$$

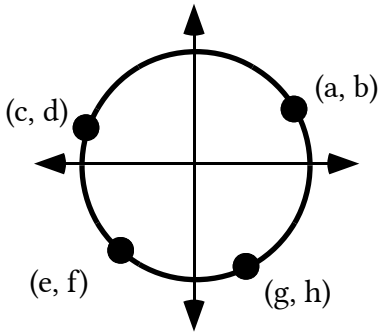
$$\left[\begin{array}{c|c} & \end{array}\right] \begin{array}{l} A_5 = \frac{-3}{2} \cdot B_4 + A_4 \\ B_5 = B_4 \end{array}$$

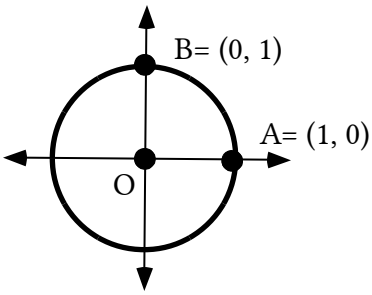

C =(4, 5)
B =(10, 2)
A

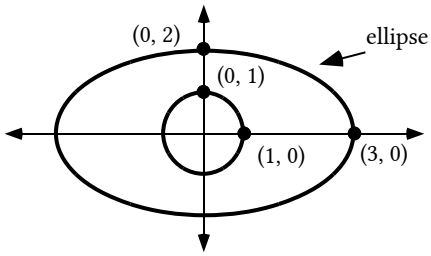
800 meter run for women

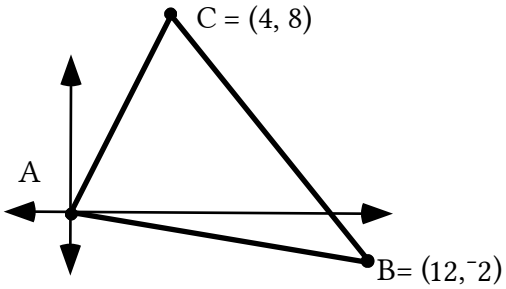
year in which world record was set	record in seconds
1953	127.3
1960	124.1
1964	121.5
1968	120.5
1971	118.3
1983	113.28

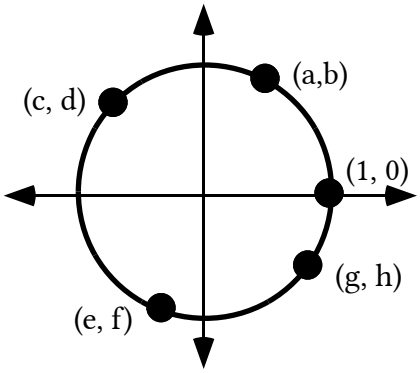

II
I
III
IV

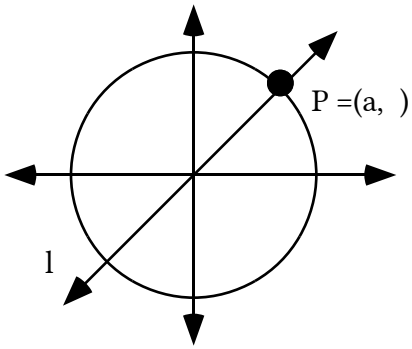

(a, b)
(c, d)
(e, f)
(g, h)

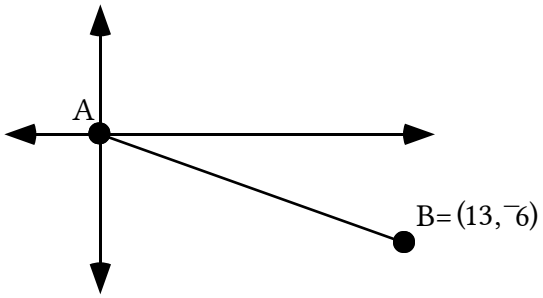

B= (0, 1)
A= (1, 0)
O

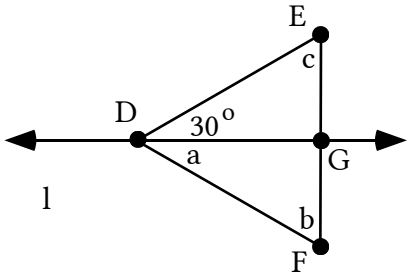

(0, 2)
ellipse
(0, 1)
(1, 0)
(3, 0)

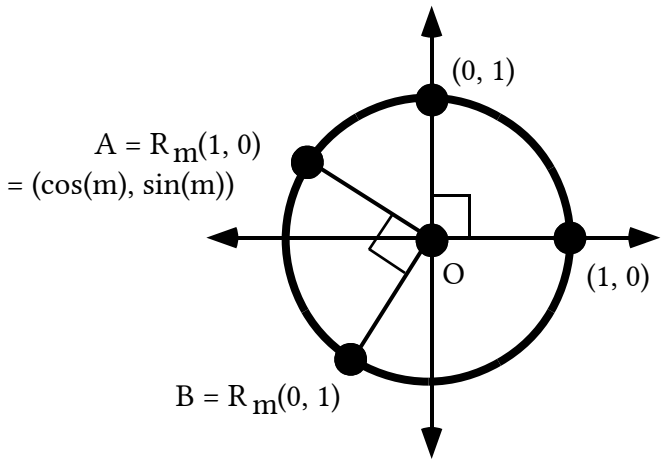

C = (4, 8)
A
B= (12,⁻2)

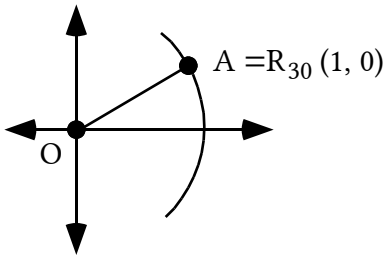

(a,b)
(c, d)
(1, 0)
(g, h)
(e, f)

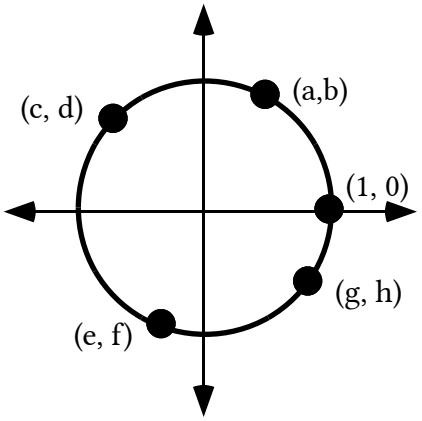

P =(a,)
l

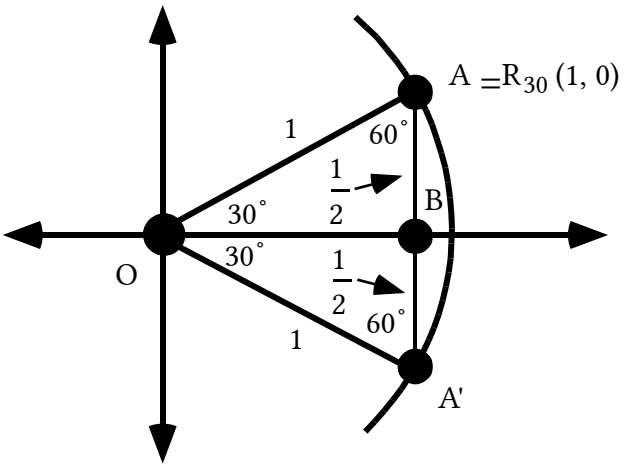

A
B= (13, ¯6)

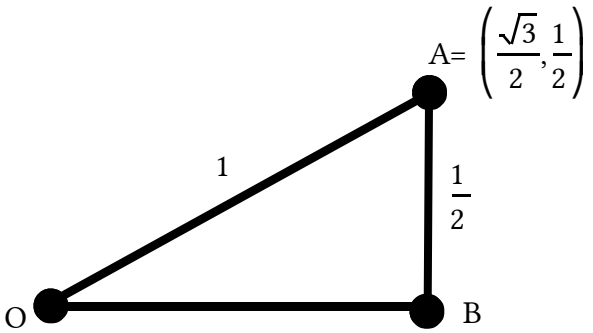

E
c
D
30 o
a
G
l
b
F


(0, 1)
A = R m(1, 0)
= (cos(m), sin(m))
O
(1, 0)
B = R m(0, 1)

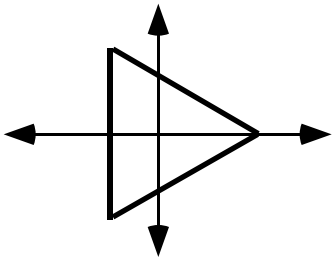

A =R30 (1, 0)
O

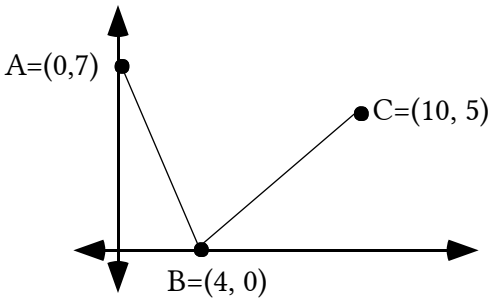

(a,b)
(c, d)
(1, 0)
(g, h)
(e, f)

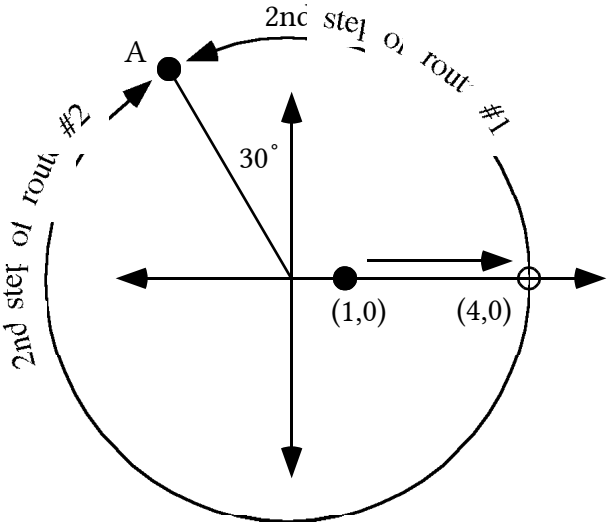

A = R_{30} (1, 0)
1
60°
1/2
B
30°
30°
O
1/2
60°
1
A'

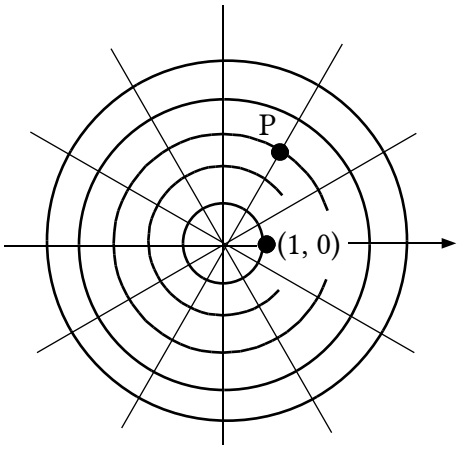


$$\sqrt{1^2-\left(\frac{1}{2}\right)^2}=\sqrt{1-\frac{1}{4}}=\sqrt{\frac{3}{4}}=\frac{\sqrt{3}}{2}$$

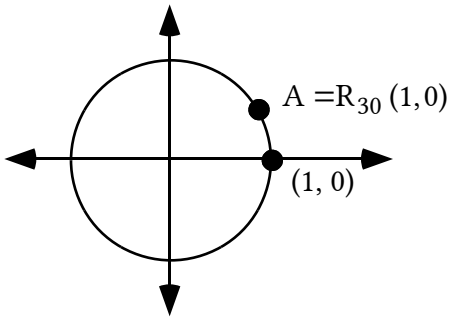



A=(0,7)
C=(10, 5)
B=(4, 0)

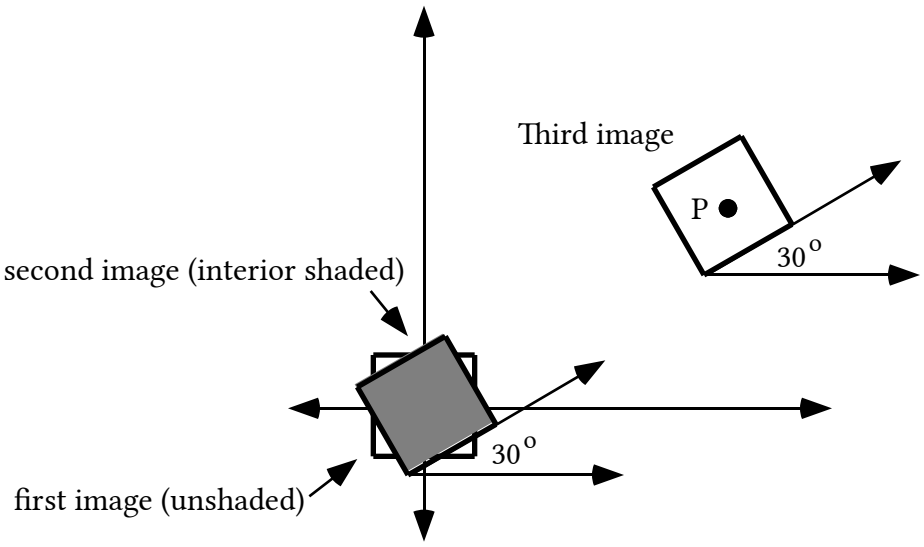

A
30°
(1,0)
(4,0)

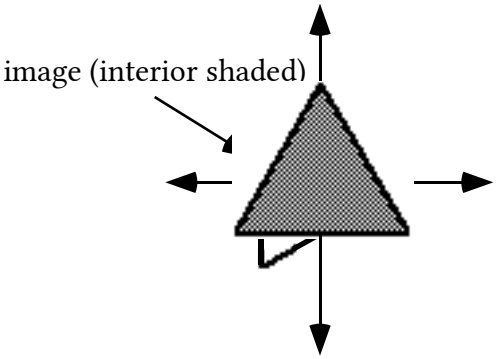

P
(1, 0)

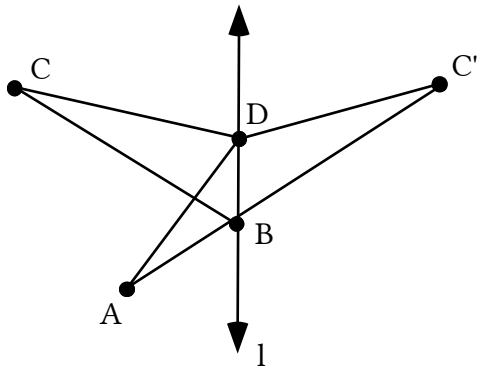
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right)$$

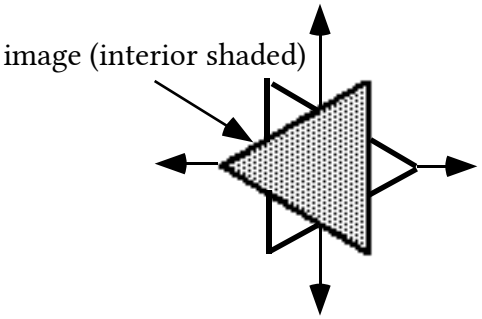

A = R_{30} (1, 0)
(1, 0)

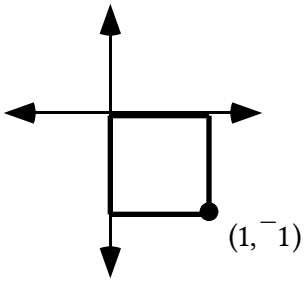
$$(\sqrt{2}/2)X_{1T} + (\sqrt{2}/2)Y_{1T}$$

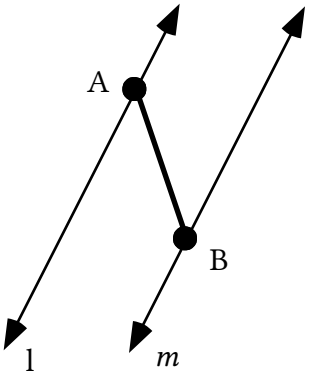

Third image
P
30°
second image (interior shaded)
30°
first image (unshaded)

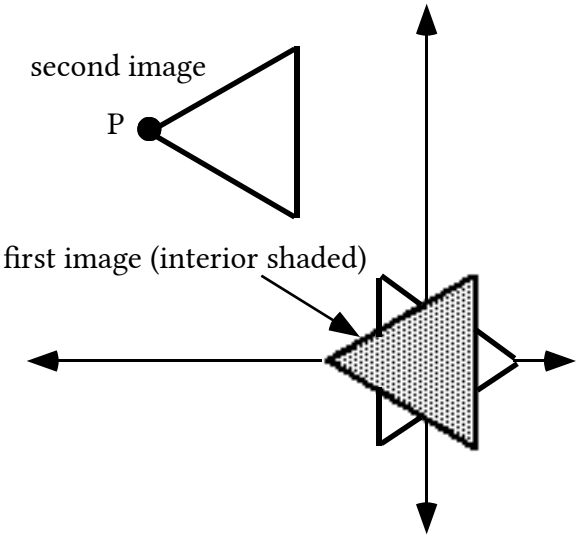

image (interior shaded)

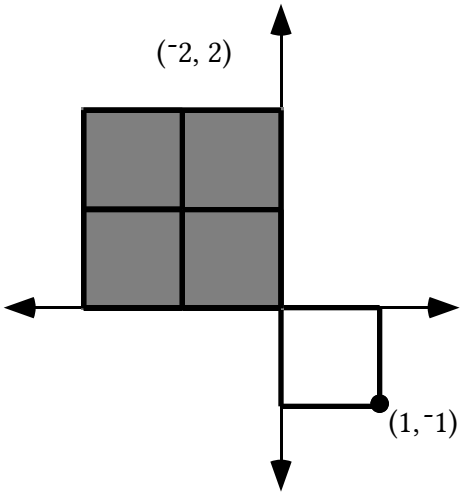

C
C'
D
B
A
l

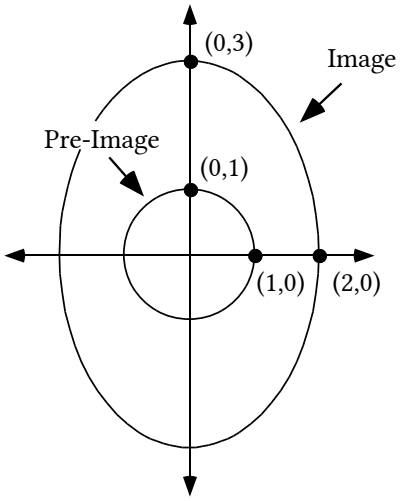

image (interior shaded)

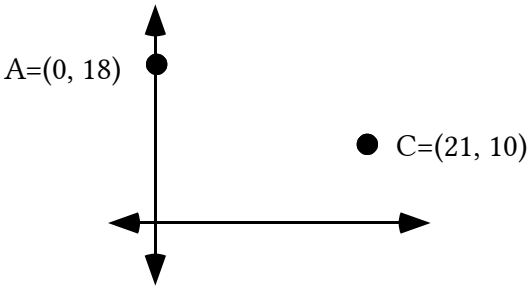

(1, −1)

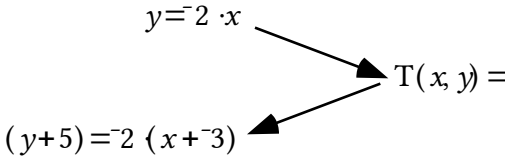

A
B
l
m

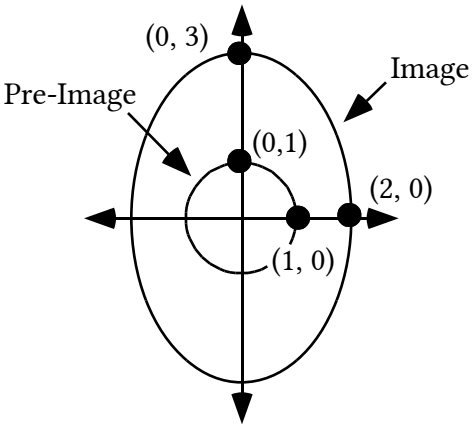

second image
P
first image (interior shaded)

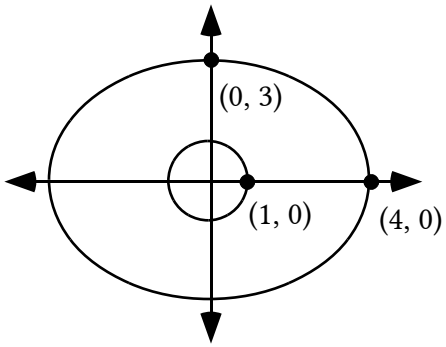

(-2, 2)
(1, -1)

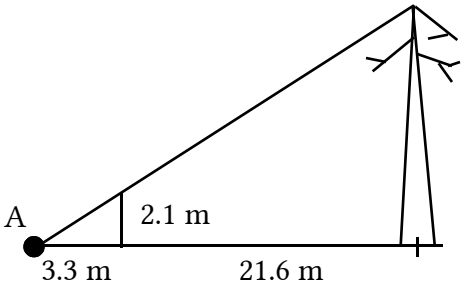

(0,3)
Image
Pre-Image
(0,1)
(1,0)
(2,0)

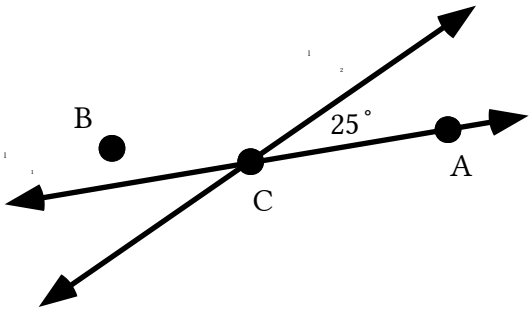

A=(0, 18)
C=(21, 10)


$y = -2 \cdot x$
$T(x, y) =$
$(y+5) = -2 \cdot (x + -3)$

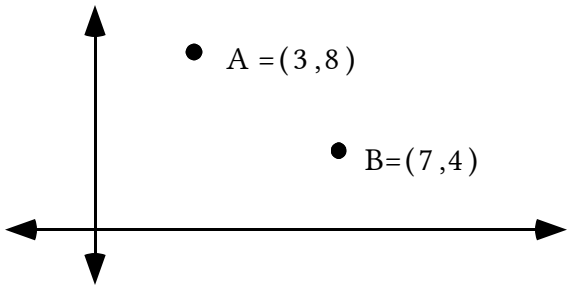

(0, 3)
Image
Pre-Image
(0,1)
(2, 0)
(1, 0)

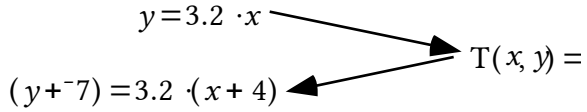

(0, 3)
(1, 0)
(4, 0)

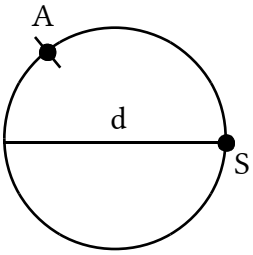

A
2.1 m
3.3 m
21.6 m

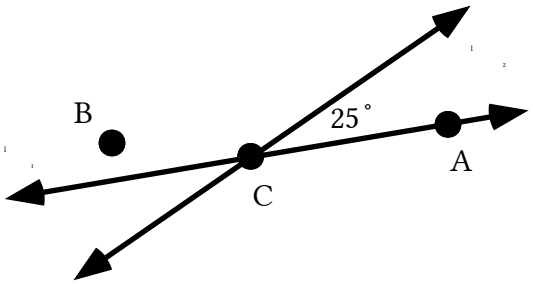

B
25˚
A
C

Team	Won	Lost
A	2	1
B	2	1
C	1	2
D	1	2

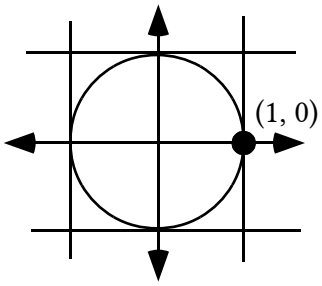

A =(3 ,8)
B=(7 ,4)

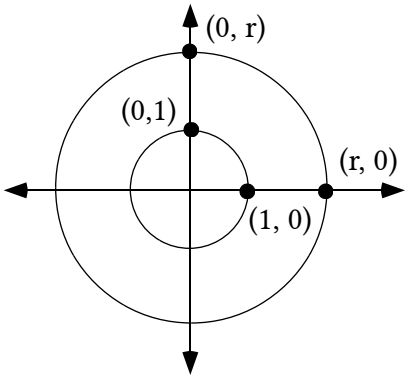

y=3.2 ·x
T(x, y) =
(y+⁻7) =3.2 ·(x+ 4)

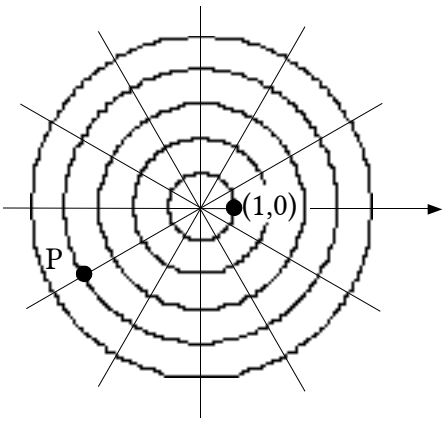

A
d
S

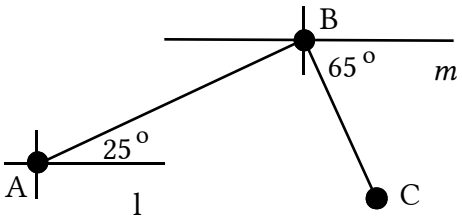

B
25˚
A
C

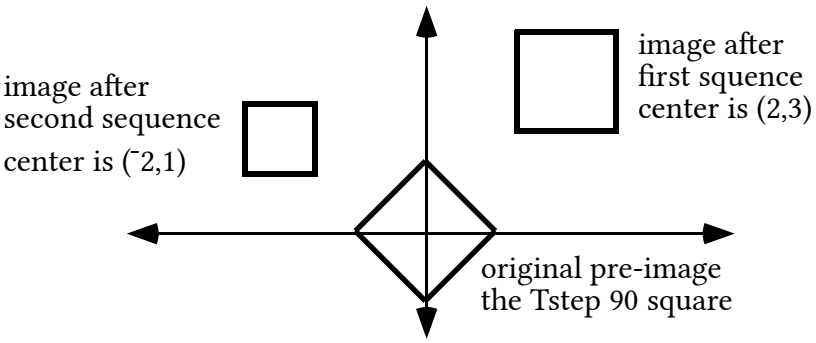
closeness index	cancer deaths
1.3	114
1.6	138
2.5	147
2.6	130
3.4	130
3.8	162
6.4	178
8.3	210
11.6	208


(1, 0)

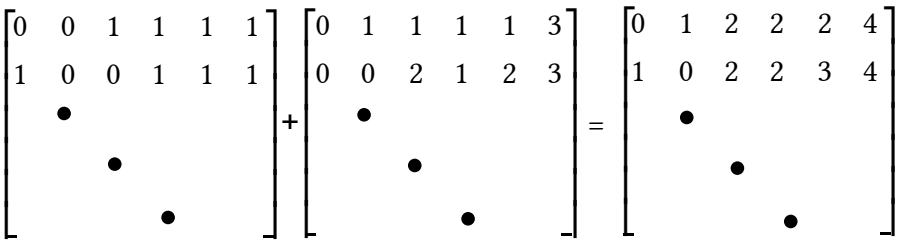

(0, r)
(0,1)
(r, 0)
(1, 0)

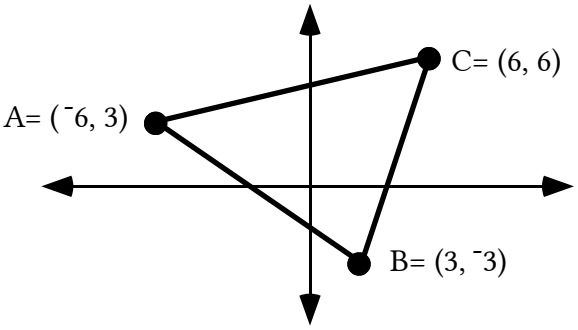

(1,0)
P


B
65 o
m
25 o
A
l
C


image after
second sequence
center is (¯2,1)
image after
first squence
center is (2,3)
original pre-image
the Tstep 90 square

addend #1 addend #2 sum




C= (6, 6)
A= (-6, 3)
B= (3, -3)